

AN IRREDUCIBLE ELEMENTARY NET OVER AN ALGEBRAIC EXTENSION THAT IS NOT CLOSED

GROUPTHEORY50

ABSTRACT. For every order $n \geq 3$ we construct an irreducible elementary net of modules over a principal ideal domain in an algebraic extension of its fraction field that is not closed. The example is defined over $\mathbb{Z}$ inside the cubic field generated by a root of $x^3 - x - 1$. This gives a negative answer to Kurovka Problem 19.48.

1. INTRODUCTION

Elementary nets, also called carpets, organize the allowed parameters of elementary transvections in matrix groups. Kurovka Problem 19.48, proposed by V. A. Koibaev, asks whether an irreducible elementary net of R -modules over an algebraic extension of the fraction field of a principal ideal domain must be closed [1]. We give an explicit counterexample.

For pairwise distinct indices i, r, j , an elementary net $\sigma = (\sigma_{ij})$ satisfies

$$\sigma_{ir}\sigma_{rj} \subseteq \sigma_{ij}.$$

It is irreducible if every off-diagonal entry is nonzero. It is closed if the elementary subgroup $E(\sigma)$ contains no transvection $x_{ij}(a)$ with $a \notin \sigma_{ij}$.

2. THE CUBIC EXAMPLE

Let α be a root of

$$\alpha^3 - \alpha - 1 = 0,$$

put $K = \mathbb{Q}(\alpha)$ and $T = \mathbb{Z}[\alpha]$. Since

$$\alpha^{-1} = \alpha^2 - 1,$$

the element α is a unit in T . Define additive subgroups

$$A = \mathbb{Z} \oplus \mathbb{Z}\alpha \oplus 2\mathbb{Z}\alpha^2,$$

$$B = \mathbb{Z} \oplus 2\mathbb{Z}\alpha \oplus \mathbb{Z}(\alpha^2 - 1), \quad C = 2T.$$

Then $C \subseteq A \cap B$ and

$$AC \subseteq C, \quad BC \subseteq C, \quad C^2 \subseteq C.$$

For $n \geq 3$, set

$$\sigma_{12} = A, \quad \sigma_{21} = B,$$

and let every other off-diagonal entry be C .

Lemma 2.1. *The family σ is an irreducible elementary net.*

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Proof. All entries are nonzero. Along a path $i \rightarrow r \rightarrow j$ with i, r, j pairwise distinct, the two exceptional entries σ_{12} and σ_{21} cannot occur consecutively. Thus every product required by the net condition is one of AC , BC , or C^2 , or is contained in an exceptional target containing C . The displayed inclusions give the claim. $\square$

3. FAILURE OF CLOSURE

In the $(1, 2)$ block write

$$x(t) = x_{12}(t), \quad y(s) = x_{21}(s),$$

and for a unit t put

$$w(t) = x(t)y(-t^{-1})x(t) = \begin{pmatrix} 0 & t \\ -t^{-1} & 0 \end{pmatrix}.$$

We have $\alpha, -1 \in A$ and $-\alpha^{-1}, 1 \in B$. Hence

$$w(\alpha)w(-1) \in E(\sigma).$$

A direct multiplication gives

$$w(\alpha)w(-1) = \text{diag}(\alpha, \alpha^{-1})$$

in the $(1, 2)$ block. Conjugating $x_{12}(1)$ by this diagonal matrix yields

$$x_{12}(\alpha^2) \in E(\sigma).$$

However $\alpha^2 \notin A$, because the coefficient of α^2 in an element of A is even.

Theorem 3.1. *For every $n \geq 3$, the elementary net σ above is irreducible but not closed.*

Proof. Irreducibility was established in Lemma 2.1, while $x_{12}(\alpha^2) \in E(\sigma)$ with $\alpha^2 \notin \sigma_{12}$ is exactly the failure of closure. $\square$

Corollary 3.2. *By Theorem 3.1, Kourovka Problem 19.48 has a negative answer, already for the PID $R = \mathbb{Z}$ and the cubic algebraic extension $K = \mathbb{Q}(\alpha)$.*

4. PERSPECTIVE

The mechanism is a rank-two diagonal element manufactured from allowed elementary transvections. Its conjugation rescales an allowed parameter outside the prescribed module. This provides a compact obstruction to closure that persists in every rank $n \geq 3$.

REFERENCES

- [1] E. I. Khukhro and V. D. Mazurov (eds.), *Unsolved Problems in Group Theory: The Kourovka Notebook*, No. 21, Novosibirsk, 2026; arXiv:1401.0300.
- [2] R. Steinberg, *Lectures on Chevalley Groups*, Yale University, 1968.